

# MATH 350-2 Advanced Calculus

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# Outline

- 1 Real Analysis Lecture 21
  - Riemann-Stieltjes integrals
  - Stieltjes integrals

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  - Stieltjes integrals

# Riemann-Stieltjes sums

Notation: given a real-valued function  $\alpha$  on  $[a, b]$

$$\Delta\alpha_k(P) = \alpha(x_k) - \alpha(x_{k-1}), \quad k = 1, 2, \dots, n.$$

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**Riemann-Stieltjes sum** is a sum of the form

$$S(P, f, \alpha, \{t_k\}) = \sum_{k=1}^n f(t_k) \Delta\alpha_k(P),$$

for some set of sample points  $\{t_k\}$  of  $P$ .

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$$|S(P, f, \alpha, \{t_k\}) - A| < \epsilon,$$

for all sets of sample points  $\{t_k\}$  of  $P$ .

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Notation: in this case  $A$  is denote by  $\int_a^b f d\alpha$  or  $\int_a^b f(x) d\alpha(x)$

# Challenge!

## Problem

Suppose that  $a, b$  are real numbers with  $a < b$  and let  $\alpha$  be a real-valued function on  $[a, b]$ .

Prove that a constant function  $f(x) = c$  is integrable with respect to  $\alpha$  on  $[a, b]$  and

$$\int_a^b f d\alpha = c\alpha(b) - c\alpha(a).$$

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## Problem

Suppose that  $a, b$  are real numbers with  $a < b$  and let  $\alpha$  be a constant function on  $a < b$ .

If  $f(x)$  is a function on  $[a, b]$ , show that  $f(x)$  is integrable with respect to  $\alpha$  on  $[a, b]$ .

What is  $\int_a^b f d\alpha$ ?

# Challenge!

## Problem

Suppose that  $a, b, c$  are real numbers with  $a < c < b$  and let  $\alpha$  be the step function

$$\alpha(x) = \begin{cases} 0, & a < x < c \\ 1, & c \leq x \leq b \end{cases}$$

Show that if  $f(x)$  is a real function on  $[a, b]$  which is continuous at  $c$ , then  $f$  is integrable with respect to  $\alpha$  on  $[a, b]$ .

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# Linearity of integration

## Theorem

*Suppose that  $f$  and  $g$  are Riemann integrable with respect to  $\alpha$  on  $[a, b]$ .*

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*Suppose that  $f$  and  $g$  are Riemann integrable with respect to  $\alpha$  on  $[a, b]$ .*

*Then for any  $c_1, c_2 \in \mathbb{R}$ ,  $c_1 f + c_2 g$  is Riemann integrable with respect to  $\alpha$  and*

$$\int_a^b (c_1 f + c_2 g) d\alpha = c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha.$$

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## Theorem

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*Then for any  $c_1, c_2 \in \mathbb{R}$ ,  $f$  is Riemann integrable with respect to  $c_1\alpha + c_2\beta$  and*

$$\int_a^b f d(c_1\alpha + c_2\beta) = c_1 \int_a^b f d\alpha + c_2 \int_a^b f d\beta.$$

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*Suppose that  $f$  is a real-valued function on  $[a, b]$  and that  $c \in (a, b)$ . If  $\int_a^c f d\alpha$  and  $\int_c^b f d\alpha$  exist, then  $\int_a^b f d\alpha$  exists and*

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

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*Conversely, if  $\int_a^b f d\alpha$  exists then  $\int_a^c f d\alpha$  and  $\int_c^b f d\alpha$  exist and*

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

# Integration by parts

The integrand and integrator play dual roles:

## Theorem

*Integration by parts* Let  $f$  be integrable with respect to  $\alpha$  on  $[a, b]$ .

Then  $\alpha$  is integrable with respect to  $f$  on  $[a, b]$  and

$$\int_a^b f d\alpha = f(b)\alpha(b) - f(a)\alpha(a) - \int_a^b \alpha df.$$

# Relation to Riemann integrals

## Theorem

*Let  $f$  be integrable with respect to  $\alpha$  on  $[a, b]$ .*

*If  $\alpha$  is continuous on  $[a, b]$  and continuously differentiable on  $(a, b)$ , then  $f(x)\alpha'(x)$  is integrable with respect to  $x$  on  $[a, b]$  and*

$$\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x) dx.$$

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# Concrete Example

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Take a partition  $P = \{x_0, \dots, x_n\}$  of  $[0, 1]$ , sample points  $\{t_k\}$ :

$$S(P, f, \alpha, \{t_k\}) = \sum_{k=1}^n f(t_k) \Delta \alpha_k(P) = \sum_{k=1}^n t_k (x_k - x_{k-1}).$$

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Since  $x$  is monotone increasing:

$$\sum_{k=1}^n x_{k-1} (x_k - x_{k-1}) \leq S(P, f, \alpha, \{t_k\}) \leq \sum_{k=1}^n x_k (x_k - x_{k-1}).$$

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Where do you go from here?

# Stieltjes sums

Let  $f$  and  $\alpha$  be real-valued functions on  $[a, b]$ , and let  $P = \{x_0, x_1, \dots, x_n\}$  be a partition of  $[a, b]$  and define

$$m_k(f) = \inf\{f(x) : x_{k-1} \leq x \leq x_k\}, \quad M_k(f) = \sup\{f(x) : x_{k-1} \leq x \leq x_k\}.$$

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The **upper Stieltjes sum** of  $f$  with respect to  $\alpha$  on  $[a, b]$  is:

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and the **lower Stieltjes sum** of  $f$  with respect to  $\alpha$  on  $[a, b]$  is:

$$\underline{S}(P, f, \alpha) = \sum_{k=1}^n m_k(f) \Delta \alpha_k(P)$$

# Stieltjes sums

These sums refine predictably!

## Lemma

*Let  $f$  and  $\alpha$  be real-valued functions on  $[a, b]$ , with  $\alpha$  monotone increasing.*

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## Lemma

*Let  $f$  and  $\alpha$  be real-valued functions on  $[a, b]$ , with  $\alpha$  monotone increasing. If  $P$  is a partition of  $[a, b]$  and  $P'$  is a refinement of  $P$ , then*

$$\underline{S}(P, f, \alpha) \leq \underline{S}(P', f, \alpha) \leq \overline{S}(P', f, \alpha) \leq \overline{S}(P, f, \alpha)$$

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# Riemann's criterion

## Definition

Let  $f$  and  $\alpha$  be real-valued functions on  $[a, b]$ .

Then  $f$  **satisfies Riemann's criterion** with respect to  $\alpha$  on  $[a, b]$  if for all  $\epsilon > 0$  there exists a partition  $P_\epsilon$  such that

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- (b)  $f$  satisfies Riemann's criterion with respect to  $\alpha$  on  $[a, b]$*
- (c)  $\underline{I}(f, \alpha) = \overline{I}(f, \alpha)$*